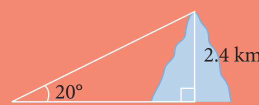
MIXED REVISION
CHAPTERS 10 • 11 • 12 4

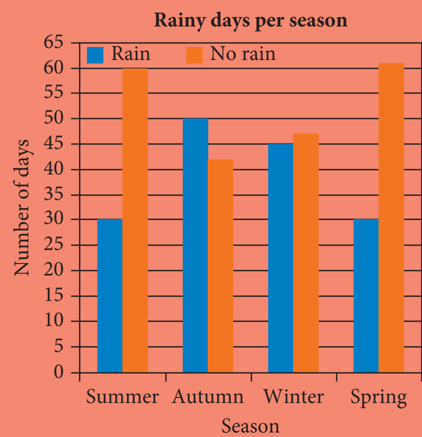
Multiple choice

1 Danielle walked up a mountain with an incline of 20° to reach a height of 2.4 km above sea level. The total distance walked by Danielle (to the nearest kilometre is) is:



A 2 km B 3 km C 5 km D 6 km E 7 km

2 The side-by-side column graph displays the number of days that it rained for each season of a year.



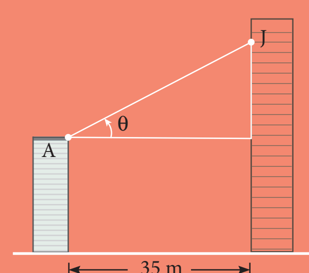
Which one of the following statements is **true**?

A There were more rainy days than dry days in all the seasons.
B Autumn had the most rainy days and the least dry days.
C Summer had more rainy days than dry days.
D Winter had the same number of rainy and dry days.
E None of the above.

3 The gradient and y-intercept respectively of the equation $3x - 4y = 20$ are

A $-3, 5$ B $\frac{3}{4}, -5$ C $\frac{4}{3}, -5$ D $\frac{3}{4}, 5$ E $5, \frac{3}{4}$

4 Angus is standing on his balcony which is 30 m above street level. He spots his friend, Josephine, on her balcony which is 60 m above street level. If the two buildings are 35 m apart, the angle of elevation of Josephine from Angus (to the nearest degree) is:

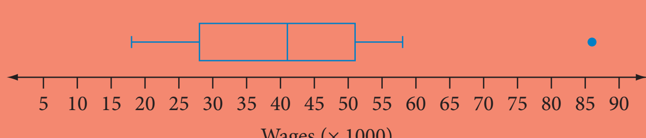


A 30° B 41° C 49°
D 60° E 90°

5 The following boxplot shows the wages of employees at a fashion outlet.

The middle 50% of staff wages is between:

A \$51 and \$28
B \$41 and \$18
C \$41 000 and \$18 000
D \$51 000 and \$28 000
E none of the above

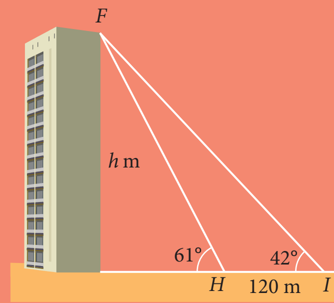


9780170250252

6 The solution to the simultaneous equations $y = 5x - 1$ and $2y = 3x - 9$ is:

A $x = -1, y = -4$ B $x = 1, y = 4$ C $x = -4, y = -21$
D $x = -1, y = -6$ E $x = 4, y = 1$

7 The length of FH can be found using:



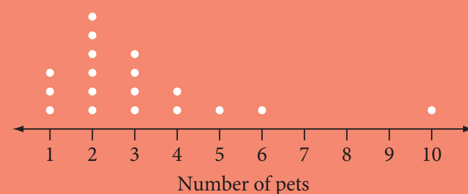
A $FH = 120 \tan (42°)$ B $FH = \dfrac{120\sin(42^\circ)}{\sin(61^\circ)}$

C $FH = \dfrac{120\sin(42^\circ)}{\sin(19^\circ)}$ D $FH = 120 \tan (61°)$

E $FH = \dfrac{120\sin(42^\circ)}{\sin(119^\circ)}$

8 The dot plot represents the amount of pocket money given to a group of 5 year old children.

Select the measure of centre and spread that best represents the data.

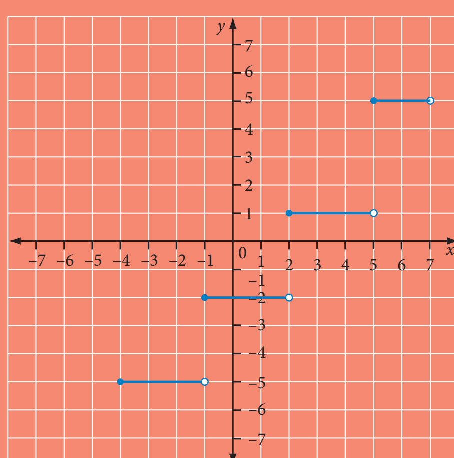


A Mode and range
B Median and interquartile range
C Mean and range
D Mean and interquartile range
E Mean and standard deviation

9 The equation which best describes the step graph is:

A $y = \begin{cases} -5, -4 \le x < -1 \\ -2, -1 \le x < 2 \\ 1, 2 \le x < 5 \\ 5, 5 \le x < 7 \end{cases}$ B $y = \begin{cases} -5, -4 < x \le -1 \\ -2, -1 < x \le 2 \\ 1, 2 < x \le 5 \\ 5, 5 < x \le 7 \end{cases}$ C $y = \begin{cases} -5, -4 \le x \le -1 \\ -2, -1 \le x \le 2 \\ 1, 2 \le x \le 5 \\ 5, 5 \le x \le 7 \end{cases}$

D $y = \begin{cases} -5, -4 \le x < -1 \\ -2, -1 \le x < 2 \\ 1, 2 \le x < 5 \\ 5, 6 \le x \le 7 \end{cases}$ E $y = \begin{cases} -5, -4 < x < -1 \\ -2, -1 < x < 2 \\ 1, 2 < x < 5 \\ 5, 6 < x < 7 \end{cases}$



Short answer questions

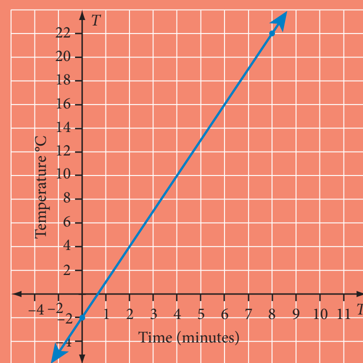
1 A triangular shade sail has sides of length 8 m, 11 m and 14 m. Calculate the area of material, correct to the nearest square metre.

2 The ages of members of a yoga class at Husky Body Shop one morning are shown below.

Women:	24	32	43	48	50	50	54	59	63	63	64	65	68		
Men:	22	26	32	32	35	36	38	42	44	45	46	53	63	63	64

a Find the interquartile range of the ages for the women in the class and interpret.
b Compare the mean age for men and women.
c Justify why no outliers exist for each gender.

3 The temperature in a room t minutes after the heating is turned on is given by the graph.
a What was the initial temperature of the room?
b By how many degrees did the temperature rise each minute?
c Use your answers to parts **a** and **b** to write a rule showing the relationship between the temperature, T, to the time, t.



4 A hang-glider pilot takes off from the top of a 100 m cliff that is at an angle of 103° to the beach. He flies 135 m to land on the beach. What is the angle of elevation from where he lands to the top of the cliff, correct to the nearest degree?

5 The following stem-and leaf plot shows the results (%) of an algebra test for Mr A's Year 10 class.

Key: 3|1 means 31%

Stem	Leaf
3	1
4	0 1 2
5	0 0 1 2 3
6	2 3 4 5 6 7 8
7	3 4 5 6 7
8	0 1 2
9	9

a Calculate the mean and standard deviation of the test results for Mr A's class.
b Why are the mean and standard deviation appropriate measures of centre and spread?

Mr A decides to compare his class' results to Mr B's class. They had an average mark of 72.61% and a standard deviation of 11.1%.

c Which teacher's class achieved higher results? Explain your reasoning.
d Which teacher's class achieved the most consistent results? Justify your thinking.

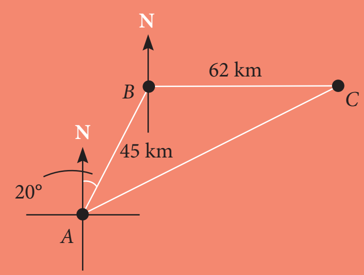
6 The equations $3x + 2y = -8$ and $7x - 3y = 35$ are to be solved simultaneously.
a Sketch the graphs of each equation on the same set of axes.
b Find their point of intersection using the graph.
c Solve the equations algebraically using substitution or elimination. Compare your answers.

9780170250252

MIXED REVISION • 10 • 11 • 12

Application questions

1 A helicopter is delivering supplies to three islands off the Australian coast as shown in the diagram below. The helicopter leaves island A on a bearing of 020° and flies for 45 km to reach island B. It then heads due east to reach island C, which is 62 km away.

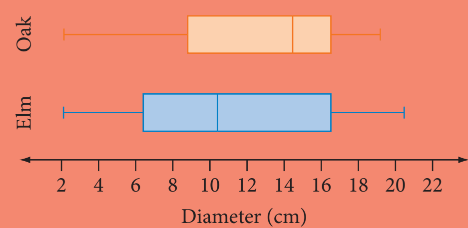


Calculate:

a The distance between island C and island A, correct to two decimal places.

b The true bearing of island A from island C, correct to the nearest degree.

2 The parallel boxplots show the diameters (cm) of a sample of elm and oak trees.



a Describe the shape of the distributions.

b Which measures of centre and spread will best represent the data?

c Explain why the mean diameter for each type of tree cannot be calculated.

d What percentage of both trees had a diameter greater than 16.5 cm?

e Which feature of the boxplot tells us that elm trees have a larger range for the middle 50% of the data?

f Compare the diameters of the two types of trees.

3 The town of WallyWater uses a piecewise linear function to calculate the monthly water charge for its consumers. The piecewise linear function, where C = the cost of the water and l = the number of thousands of litres, is:

$$C = \begin{cases} 5l + 20, & 0 \le l \le 5 \\ 9l, & 5 < l \le 10 \\ 10l - 10, & l > 10 \end{cases}$$

a Sketch the graph of the function to show the relationship between the litres used and the cost of water in WallyWater town.

b Describe the graph you have sketched.

c Which points are the critical points on the graph. Explain your reasoning.

d What do you notice about the slope of each line segment? What do you think the changing slope means?

e Use your graph to determine the monthly water charge if a consumer used:

i 3500 litres   ii 9000 litres   iii 12 500 litres

9780170250252